

ment of applications for a variety of uses. It strategically mixes flexibility and security while also providing enterprise-grade features, making it a popular choice among many companies.

Of course, the very first step is to understand the fundamentals of Ethereum, such as its definition, functional scope, and main strengths and limits. Ethereum offers smart contract execution as a decentralised platform. Smart contracts are computer programs that defend against fraudulent operations, downtimes, censorship, external meddling, and other attempts to change or modify the terms of a contract.

Smart contracts are made possible by a custom designed blockchain that runs on globally shared infrastructure with enormous capacities. Ethereum includes robust capabilities for reflecting property ownership, in addition to frictionless and decentralised transactions—two aspects that make it a viable solution for both commercial and institutional entities. Ethereum is a software framework that allows developers to create sophisticated decentralised apps with cutting-edge features.

If you are wondering how much an Ethereum developer can earn, it depends on factors like expertise, necessary qualifications, the size/business of the employer, location, and so on. According to ZipRecruiter, an Ethereum developer with sufficient experience might make up to \$172,000 per year. You can also look for work as a freelance Ethereum developer, which pays \$70-100 per hour.

Requirements for becoming an Ethereum developer

To become an Ethereum blockchain developer, your knowledge of blockchain technology would have to be thorough. It should not be limited to some often used phrases. You will need a complete understanding of how things function behind

the scenes, as well as the logical relationship between back-end procedures and front-end results, blockchain operation and capabilities, real-world applications, and current development issues.

To be more specific, you must have a good understanding of:

- Smart contracts
- Consensus algorithms
- Transactions
- Gas/Gas prices
- Miners' incentives and security
- Sharding
- Proof of work/Proof of stake comparison
- Token standards

Ethereum's main goal is to provide secure public distribution; cryptography is at the heart of this technology. As a result, aspiring Ethereum engineers must have a thorough understanding of cryptography, including how it works, how it is implemented in practice, and what its possibilities and limits are.

Many people believe that cryptography is only a small portion of the blockchain development process. It is, however, separated into several sub-topics, each of which must be learned separately. Here are some significant cryptography topics to become acquainted with:

- Public key encryption
- Private key encryption
- Digital signatures
- Hash functions
- Key exchange or agreement
- Elliptic curve encryption
- Encrypted storage and ring signatures
- Trusted execution environments

• Proofs for zero-knowledge
Buyers and sellers enter into self-executing agreement contracts known as smart contracts. The agreement is written in binary code. It runs on the blockchain network and is decentralised. Smart contracts provide blockchain networks with enterprise-grade logic, making them more desirable to organisations. As

a result, most blockchain networks are incorporating smart contract capability into their platforms as quickly as possible.

You will therefore require a thorough understanding of smart contract development, covering main languages like Chaincode, Viper, Solidity, and other relevant languages.

Programming and web development

Programming languages are crucial in the development of diverse Ethereum solutions for varied use cases. To begin learning Ethereum blockchain development, you will require exceptional programming abilities. The pace and intensity of your Ethereum development learning will be strongly influenced by your programming knowledge.

It is worth noting that Solidity is the Ethereum ecosystem's most popular programming language, and the best part is that it is a lot easier to learn and operate with due to its grammatical similarities to JavaScript. Here is a rundown of some of the most important programming languages to know:

- Python
- Ruby
- JavaScript
- Java
- C++
- Go
- Rust
- .NET

Many blockchain development opportunities for entrants are in the field of designing various basic decentralised applications. Web development is an integral aspect of blockchain development. So, you need decent knowledge of how front-end and back-end development works. To be more specific, you need to have good knowledge about request handling, API handling, and GUI creation for Dapps. **EFY**

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